

EME-401: Automation and Control Systems

Lecture 1: Introduction to Modern Automation and Control

Instructor: Dr. Mustafa S. Aljumaily

Fall 2025

Course Objectives

-  **Analyze and Model:** Develop mathematical models for various physical systems (mechanical, electrical, electromechanical) relevant to automation and control.
-  **Design and Implement:** Design and implement classical (e.g., PID) and modern control strategies for real-world electromechanical applications.
-  **Utilize Software Tools:** Effectively use software tools (e.g., MATLAB/Simulink) for control system simulation, analysis, and design.
-  **Understand Advanced Concepts:** Grasp the fundamentals of state-space analysis and digital control.
-  **Integrate Automation Systems:** Understand the selection, integration, and role of sensors, actuators, and PLCs in complete industrial automation systems.
-  **Evaluate and Troubleshoot:** Analyze the performance of control systems, identify common issues, and apply troubleshooting techniques.
-  **Recognize Modern Trends:** Identify and discuss emerging trends and technologies in modern automation and control, such as AI integration and smart systems.

Course Structure and Assessment

Course Modules

Module 1: Fundamentals of Control Systems (Weeks 1-3)

System modeling, transfer functions, time/frequency domain analysis, stability

Module 2: Classical Control Design (Weeks 4-6)

Root locus, frequency response, PID controllers, digital implementation

Module 3: Modern Control Techniques (Weeks 7-9)

State-space representation, controllability, observability, digital control

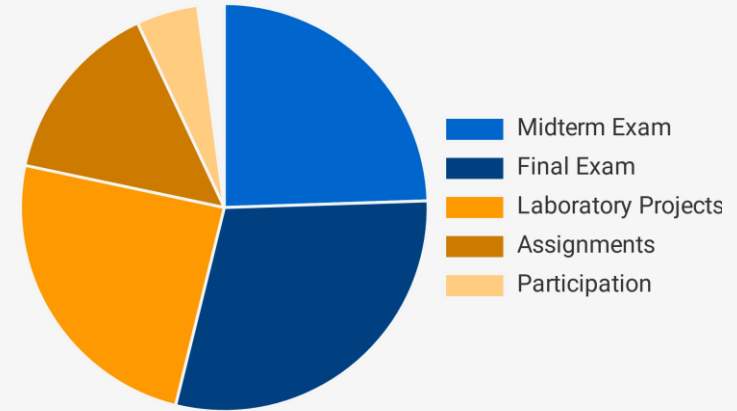
Module 4: Industrial Automation Components (Weeks 10-12)

Sensors, actuators, PLCs, HMI systems

Module 5: System Integration and Modern Trends (Weeks 13-15)

Industrial protocols, system integration, Industry 4.0, IIoT

Assessment Methods



Required Materials

Nise, N. S. (2023). Control Systems Engineering (8th ed.). Wiley.

MATLAB/Simulink (student version)

Laboratory equipment access

What is Automation and Control?

Automation: The use of control systems for operating equipment with minimal human intervention.

Control: The regulation of system behavior to achieve desired outcomes through the manipulation of inputs based on measured outputs.

Key Concepts

↔ **Open-loop vs. Closed-loop:** Open-loop systems operate without feedback, while closed-loop systems use feedback to adjust control actions.

↻ **Feedback Principle:** Comparing desired output with actual output to minimize error and improve system performance.

⌋ **System Types:** Continuous vs. discrete, linear vs. nonlinear, SISO (Single Input, Single Output) vs. MIMO (Multiple Input, Multiple Output).



Modern factory automation system showcasing integrated control and robotics

Control Systems Fundamentals

Control System Types

→ Open-Loop Control

System where output has no influence on the control action. No feedback or error correction mechanism exists.

↻ Closed-Loop (Feedback) Control

System where output is measured and fed back to compare with reference input. Automatically adjusts to minimize error.

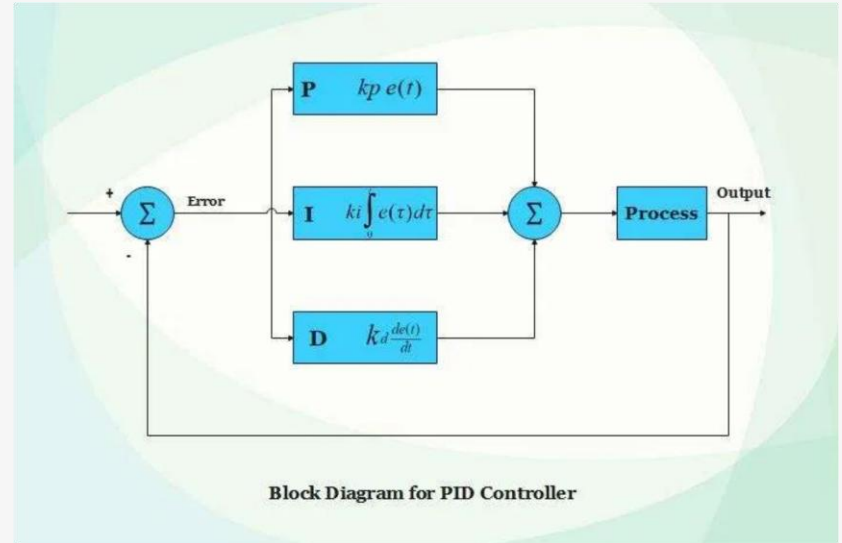
Control System Classifications

⌚ Continuous vs. Discrete

Continuous systems process analog signals, while discrete systems operate on sampled data at specific time intervals.

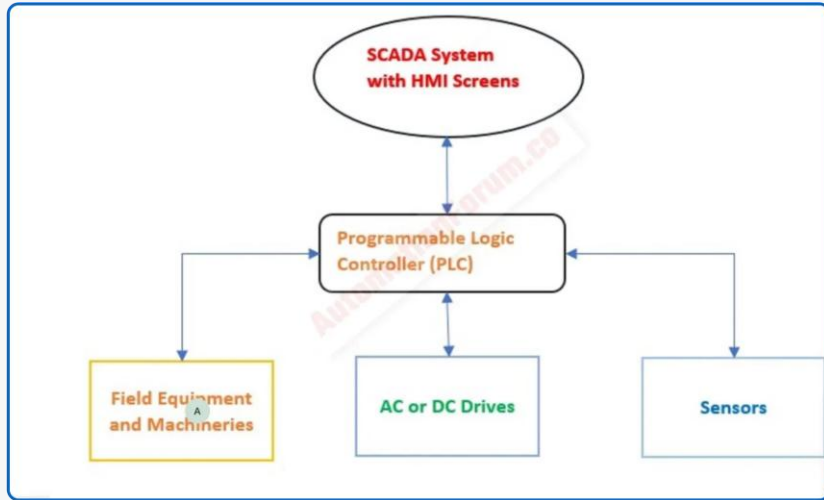
🔗 SISO vs. MIMO

Single-Input-Single-Output (SISO) vs. Multiple-Input-Multiple-Output (MIMO) systems, with increasing complexity.



PID Controller Block Diagram showing feedback control structure with proportional, integral, and derivative components

Components of Modern Control Systems



Modern industrial automation system architecture showing key components and their interconnections

🔌 Sensors

Measure physical quantities (position, temperature, pressure) and convert them to electrical signals with varying accuracy, resolution, and response time.

⚙️ Controllers

Process input signals and determine control actions, including PID controllers, PLCs, and embedded microcontrollers that execute control algorithms.

⚡ Actuators

Convert controller signals into physical actions, such as electric motors, hydraulic/pneumatic systems, and solenoids that manipulate the controlled system.

📡 Communication Systems

Enable data exchange between components using industrial protocols (Modbus, Profibus, EtherCAT) and interfaces that connect the system with operators.

Industrial Applications

Manufacturing Automation

Assembly lines and work cells

Quality control systems

Material handling and logistics

Process Control

Continuous process monitoring

Batch processing systems

Chemical and pharmaceutical production

Building Automation

HVAC control systems

Security and access control

Energy management systems

Transportation Systems

Automotive control systems

Railway signaling and control



Modern manufacturing facility with robotic automation systems

Modern Trends in Automation (2025)

Industrial Internet of Things (IIoT)

Connected sensors and devices enabling real-time data collection, analysis, and system optimization.

Edge Computing

Processing data closer to the source for faster response times in critical control applications.

AI and Machine Learning

Predictive maintenance, adaptive control strategies, and process optimization through intelligent algorithms.

Human-Robot Collaboration

Advanced safety systems and intuitive interfaces enabling humans and robots to work together efficiently.

Digital Twins

Virtual replicas of physical systems for simulation, testing, and optimization before implementation.



Modern robotic manufacturing line with advanced automation



Industrial automation system with robotic arms and control displays

Course Roadmap

Theory vs. Practical Focus by Module



Fundamentals

Master core control theory concepts and mathematical foundations



Implementation

Apply theory to practical control system design and hardware integration



Simulation

Use software tools to model and analyze control systems before physical implementation



Integration

Combine components into complete automation solutions and system testing

Next Steps

Upcoming Lecture

Lecture 2: System Modeling and Transfer Functions

- Mathematical modeling of physical systems
- Mechanical and electrical system analogies
- Transfer function derivation
- Block diagram representation

Assignments

Reading Assignment

Chapter 1 of Nise, N. S. (2023). Control Systems Engineering (8th ed.). Wiley.

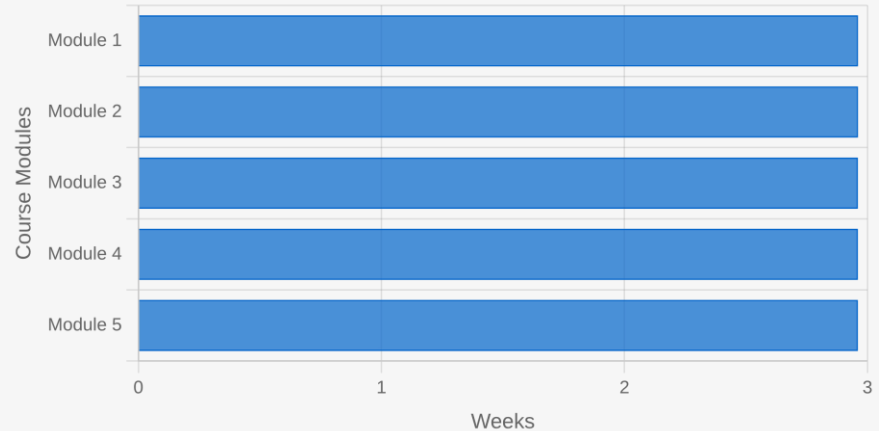
Reflection Paper

Write a brief reflection (1-2 pages) on automation in your intended career path.

Pre-Lab Preparation

Review the Lab 1 manual and complete the pre-lab questions.

Course Timeline



Contact Information

 Office Hours: Mondays and Wednesdays, 2:00 PM - 4:00 PM

 Email: professor@university.edu

 Office: Engineering Building, Room 305

 Discussion Forum: Available on the course LMS